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$$i\hbar \frac{d}{dt} |\psi(t)\rangle = E_0 \alpha_0 e^{-2i\omega_0 t} |E_0\rangle + E_1 \alpha_1 e^{-2i\omega_1 t} |E_1\rangle$$

- WE INTRODUCE THE **ENERGY OPERATOR H** (AKA **HAMILTONIAN**)
- $H|E_n\rangle = E_n|E_n\rangle$ FOR AN ENERGY LEVEL STATE $|E_n\rangle$
- H ACTS IN A LINEAR WAY
- $\hookrightarrow$ H IS NOT UNITARY!

$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$ THIS IS THE SO CALLED
SCHRÖDINGER'S EQUATION + COMP

EVOLUTION OF A SPIN-1/2 PARTICLE

FROM A SCHRÖDINGER'S EQUATION, IT FOLLOWS THAT **ENERGY** CONTROLS THE EVOLUTION OF A SYSTEM

$$|\psi(0)\rangle = \alpha_+ |z_+\rangle + \alpha_- |z_-\rangle \quad |\psi(t)\rangle = ?$$

THE ENERGY OF THE PARTICLE (DUE TO ITS SPIN) DEPENDS ON THE MAGNETIC FIELD B

$$E = -B\mu_z = -\gamma B S_z$$

$\hookrightarrow$ IS THE VALUE OF THE SPIN COMPONENT ALONG Z
 $\hookrightarrow$ **GYROMAGNETIC**

CAN WE WRITE DOWN THE HAMILTONIAN OPERATOR?

• ANY **OBSERVABLE** QUANTITY CAN BE ASSOCIATED WITH AN **OPERATOR** ACTING ON KETS

• WE SAW THAT WE CAN DEFINE THE OPERATOR S_z ASSOCIATED WITH THE MEASUREMENT OF THE SPIN

WE CAN WRITE THE HAMILTONIAN OPERATOR STARTING FROM THE SPIN OPERATOR

$$H = -\gamma B S_z$$

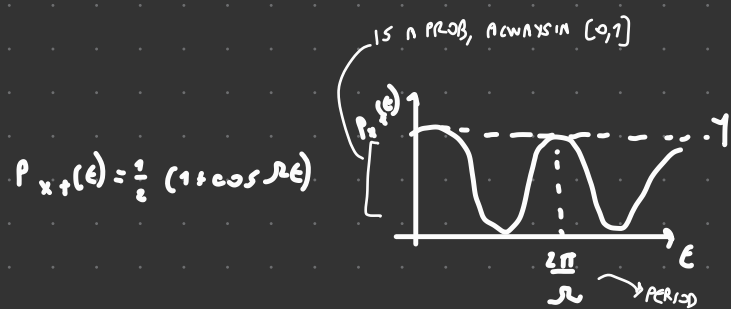
$$H |z_+\rangle = +\gamma B \hbar/2$$

$$E_+ = -\gamma B \hbar/2 \quad E_- = +\gamma B \hbar/2$$

WE CAN DEFINE **LARMOR FREQUENCY** $\Omega = \gamma B$

SOME NUMER (PROTON)

$$|\psi(0)\rangle = |x_+\rangle = \frac{1}{\sqrt{2}} |z_+\rangle + \frac{1}{\sqrt{2}} |z_-\rangle \rightarrow |\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{i\Omega t/2} |z_+\rangle + \frac{1}{\sqrt{2}} e^{-i\Omega t/2} |z_-\rangle$$



THE STATE WILL OSCILLATE BETWEEN $|x_+\rangle$ AND $|x_-\rangle$ WITH A FREQUENCY Ω

ISOMORPHISM

PHOTONS
"UPPER" _____
"LOWER" _____

$$|0\rangle = |\text{lower}\rangle$$

$$|1\rangle = |\text{upper}\rangle$$

SPIN-1/2 PARTICLES

"UP" $\uparrow$
"DOWN" $\downarrow$

$$|0\rangle = |\uparrow\rangle$$

$$|1\rangle = |\downarrow\rangle$$

TWO LEVEL ATOMS

E_1 _____
 E_0 _____

$$|0\rangle = |E_0\rangle$$

$$|1\rangle = |E_1\rangle$$

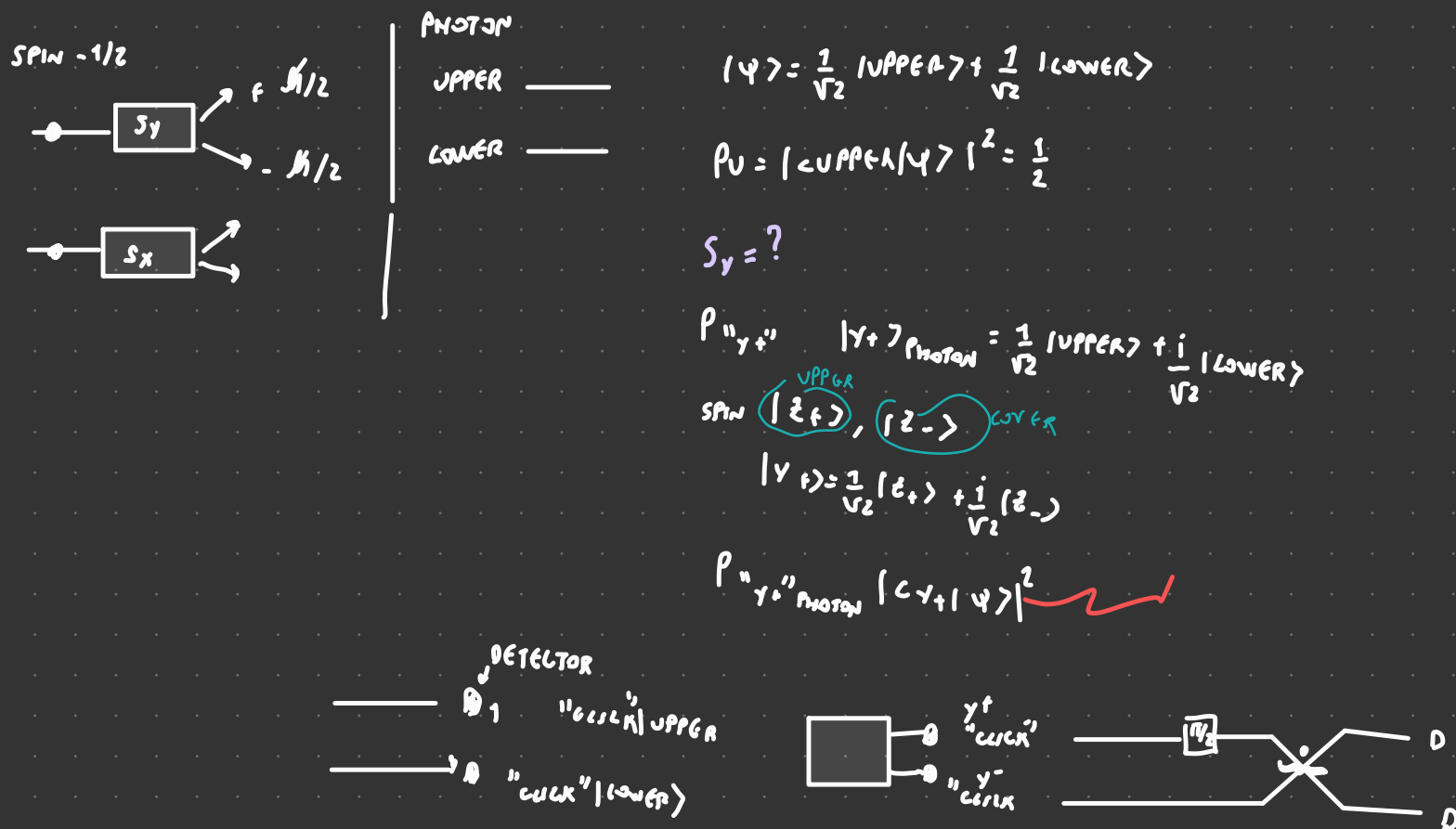
$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$$

IN ANALOGY WITH CLASSICAL INFORMATION THEORY WE TALK ABOUT A **QUBIT** STATE

QUBIT (SUMMARY) $|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$

- WE CAN ALWAYS WRITE A QUBIT AS A COMPLEX LINEAR SUPERPOSITION OF THE STATES $|0\rangle$ & $|1\rangle$
- THE STATES $|0\rangle$ & $|1\rangle$ FORM AN ORTHONORMAL BASIS FOR THE VECTOR SPACE (IE. $\langle 1|0\rangle = 0$ AND $\langle 0|0\rangle = \langle 1|1\rangle = 1$) SOMETIMES THIS BASIS IS CALLED COMPUTATIONAL
- THE COEFFICIENTS $\alpha_k = \langle k|\psi\rangle$ ARE THE PROBABILITY AMPLITUDES FOR THE TWO POSSIBLE

EXERCISE: DEVISE AN APPARATUS TO MEASURE THE ANALOG OF S_y IN THE TWO-BEAM INTERFEROMETER



THE PROTOTYPE QUBIT SYSTEM OF THE LAST CHAPTER ARE VERY SIMPLE, BUT THEY CAN BE GENERALIZED TO MORE COMPLICATED VERSION

1. WE NEED A **GENERAL** MATHEMATICAL STRUCTURE THAT IS APPLICABLE TO MANY KINDS OF SYSTEM
2. WE NEED TO DESCRIBE HOW TO APPLY THE QUANTUM FORMALISM TO **SPECIFIC PHYSICAL** SITUATIONS

MAIN POINTS

- THE STATES OF A QUANTUM SYSTEM ARE DESCRIBED BY KETS $|\psi\rangle$, WHICH OBEY THE PRINCIPLE OF SUPERPOSITION
- THE KETS ARE ELEMENTS OF AN **ABSTRACT** VECTOR SPACE H CALLED A **HILBERT SPACE**

PROPERTY OF A HILBERT SPACE H ($H = \text{HAMILTONIAN}$)

1. IT IS VECTOR SPACE WITH COMPLEX COEFFICIENTS

$$\begin{aligned} &|\psi\rangle, |\phi\rangle, |\xi\rangle \\ &|\psi_1\rangle, |\psi_2\rangle \\ &|\psi_3\rangle = \alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle \end{aligned}$$

$\alpha_1, \alpha_2 \dots \in \mathbb{C}$ COMPLEX NUMBER

2. IT HAS INNER PRODUCT

FUNCTIONAL

$$|\psi\rangle, |\phi\rangle \in \mathcal{H} \rightarrow \langle \psi | \phi \rangle \in \mathbb{C}$$

2.A SYMMETRY $\langle \psi | \phi \rangle = \langle \phi | \psi \rangle^*$

2.B LINEARITY

$$\langle \psi | (\alpha_1 |\phi_1\rangle + \alpha_2 |\phi_2\rangle) = \alpha_1 \langle \psi | \phi_1 \rangle + \alpha_2 \langle \psi | \phi_2 \rangle$$

2.C POSITIVE-DEFINITENESS

$$\langle \psi | \psi \rangle \in \mathbb{R}^+ \text{ (including 0)}$$

3 NORM OF $|\psi\rangle$ $||\psi|| = \sqrt{\langle \psi | \psi \rangle}$

→ VECTOR OF A HILBERT SPACE CAN BE DESCRIBED USING A SET OF PARTICULAR VECTORS. THIS SET IS CALLED **BASIS**

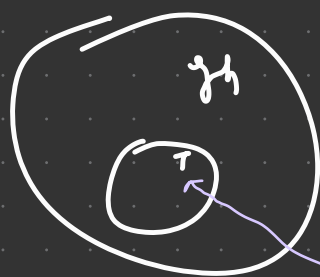
$|\psi\rangle = \sum_N \underbrace{\langle \psi | \phi_N \rangle}_{\text{component}} \underbrace{|\phi_N\rangle}_{\text{basis vector}}$ IF $\{|\phi_N\rangle\}$ IS A BASIS, ANY VECTOR OF THE HILBERT SPACE CAN BE WRITTEN LIKE THIS

• THE BASIS IS NOT UNIQUE, I CAN FIND A BASIS OF VECTORS $\{|\epsilon_N\rangle\}$

$$|\psi\rangle = \sum_N c_N |\epsilon_N\rangle$$

• ALL THE BASIS HAVE THE SAME NUMBER OF VECTORS THIS NUMBER IS CALLED **DIMENSION OF THE HILBERT SPACE**

$\dim \mathcal{H}$



$T = \text{SUBSPACE OF } \mathcal{H}$

A SUBSPACE IS A SUBSET OF A HILBERT SPACE, BUT IT IS STILL AN HILBERT SPACE

$$\dim T \leq \dim \mathcal{H}$$

E.g. $\mathcal{H} = \{|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, |\phi_4\rangle\}$

$$\mathcal{B}_T = \{|\phi_1\rangle, |\phi_3\rangle\} \quad |\psi\rangle = \alpha |\phi_1\rangle + \beta |\phi_3\rangle \in T$$

$$|\phi_1\rangle \in \mathcal{H} \quad |\phi_2\rangle \notin T$$

AN ORTHONORMAL BASIS IS SUCH THAT

$$\forall m, n \quad \langle \phi_m | \phi_n \rangle = \delta_{mn} \quad \delta_{mn} = \begin{cases} 1 & m=n \\ 0 & m \neq n \end{cases}$$

$$|\phi_N\rangle \text{ ORTHOGONAL TO } |\phi_M\rangle \Leftrightarrow \langle \phi_N | \phi_M \rangle = 0$$

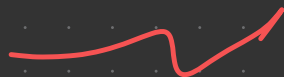
$$|\alpha\rangle = \sum_N a_N |\phi_N\rangle$$

$$|\beta\rangle = \sum_N b_N |\phi_N\rangle$$

$$\longrightarrow \langle \alpha | \beta \rangle = \sum_N a_N^* b_N$$

$$\langle \alpha | \beta \rangle = \langle \alpha | \sum_N b_N |\phi_N\rangle = \sum_N b_N \langle \alpha | \phi_N \rangle$$

$$= \sum_N b_N \langle \phi_N | \alpha \rangle^* = \sum_N b_N (\langle \phi_N | \sum_M a_M |\phi_M\rangle)^*$$



• THE STATE OF A QUANTUM SYSTEM ARE DESCRIBED BY KETS $|\psi\rangle$ WHICH OBEY THE PRINCIPLE OF SUPERPOSITION

• THE KETS ARE ELEMENTS OF AN **ABSTRACT** VECTOR SPACE H CALLED A **HILBERT SPACE**

EXERCISE: SHOW THAT IF $|\alpha\rangle = \sum_N \omega_N |\phi_N\rangle$ FOR AN ORTHONORMAL BASIS $\{|\phi_N\rangle\}$ THEN $\omega_M = \langle \phi_M | \alpha \rangle$

$$\langle \phi_M | \alpha \rangle = \langle \phi_M | \left(\sum_N \omega_N |\phi_N\rangle \right) \rangle \stackrel{\text{LINEARITY}}{=} \sum_N \omega_N \langle \phi_M | \phi_N \rangle = \sum_N \omega_N \delta_{MN} = \omega_M$$

EXERCISE: SHOW THAT, IF THE STATE $|\psi\rangle = |\phi_M\rangle$ FOR SOME BASIS VECTOR $|\phi_M\rangle$, THE MEASUREMENT WILL YIELD THE M -TH RESULT WITH CERTAINTY

$$P_K(|\psi\rangle) = |\langle K | \psi \rangle|^2 = |\langle K | \phi_M \rangle|^2 = |\delta_{KM}|^2 \quad \begin{aligned} &\text{IF } K=M \text{ THEN } P_M(|\psi\rangle) = 1 \\ &\text{IF } K \neq M \text{ THEN } P_K(|\psi\rangle) = 0 \end{aligned}$$

EXERCISE SHOW THAT, FOR ANY QUANTUM STATE $|\psi\rangle$ AND ANY BASIC MEASUREMENT THE PROBABILITIES SATISFY THE USUAL LAWS:

$$\begin{aligned} P(K) &\geq 0 \\ \sum_K P(K) &= 1 \end{aligned}$$

$$P_K(|\psi\rangle) = |\langle K | \psi \rangle|^2$$

↳ ANYTHING

$$|H\rangle = \sum_N K_N |\phi_N\rangle \quad \begin{aligned} &\text{WE ASSUME THERE IS A BASIS} \\ \langle K | H \rangle &= 1 = \sum_N |K_N|^2 = 1 \end{aligned}$$

$$|\psi\rangle = \sum_m \psi_m |\phi_m\rangle \quad \langle \psi | \psi \rangle = 1 = \sum_n |\psi_n|^2 = 1$$

$$\begin{aligned} \langle H | K \rangle &= \sum_N \langle K | K_N |\phi_N\rangle = \sum_N K_N \langle K | \phi_N \rangle = \sum_N K_N \delta_{KN} = \sum_N K_N \langle \phi_N | \phi_K \rangle^* = \sum_N K_N \delta_{KN} \\ &= \sum_N \sum_M K_N K_M^* \delta_{MN} = \text{wavy line} \end{aligned}$$

$$P_K(|\psi\rangle) = |\langle K | \psi \rangle|^2 = |\langle K | \sum_n \psi_n |\phi_n\rangle|^2 = \left| \sum_n \psi_n \langle K | \phi_n \rangle \right|^2 = \left| \sum_n \psi_n \delta_{Kn} \right|^2 = \left| \sum_n \psi_n \right|^2$$

$$= \left| \sum_n \psi_n \delta_{Kn} \right|^2 = \left| \sum_n \psi_n \left(\sum_m K_m^* \delta_{nm} \right) \right|^2 = \left| \sum_m \psi_m \sum_n K_m^* \delta_{nm} \right|^2 = \left| \sum_m \psi_m K_m^* \right|^2$$

$$= \left| \sum_m \psi_m K_m^* \right|^2$$

$$|K\rangle = |\phi_K\rangle \quad \text{BASIC MEASUREMENT} \quad P(K) = |\psi_K|^2 \leq 1$$

$$\sum P(K) = \sum |\psi_K|^2 = 1$$

KETS, BRA AND MATRICES

A GOOD WAY TO DO CALCULATIONS WITH KETS IS TO USE MATRICES.
SUPPOSE WE FIX AN ORTHONORMAL BASIS IN A D-DIMENSIONAL HILBERT SPACE $\mathcal{H}$ THE

$$\langle \alpha | \beta \rangle = \alpha^\dagger \beta = (\alpha_1^* \dots \alpha_d^*) \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_d \end{pmatrix}$$

$\langle \alpha |$ IS ALSO CALLED DUAL VECTOR (LINEAR FUNCTION)

$$|\beta\rangle \longrightarrow \langle \alpha | \beta \rangle \in \mathbb{C}$$

COMPLEX NUMBER

	KET	BRA
	$ \alpha\rangle = \sum_{n=1}^d \alpha_n \phi_n\rangle$	$\langle \alpha = \sum_{n=1}^d \alpha_n^* \langle \phi_n $
BASIS	$\{ \phi_n\rangle \}$	$\{ \langle \phi_n \}$
COMPONENT	$\alpha = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_d \end{pmatrix}$	$\alpha^\dagger = (\alpha_1^* \dots \alpha_d^*)$

NOTE THAT EVERY DUAL VECTOR ON $\mathcal{H}$ IS THE "BRA" OF SOME "KETS"

AN OPERATOR G ON THE HILBERT SPACE $\mathcal{H}$ IS A FUNCTION THAT MAPS VECTORS TO VECTORS IN A LINEAR WAY

$$G(\alpha_1 |\phi_1\rangle + \alpha_2 |\phi_2\rangle) = \alpha_1 G(|\phi_1\rangle) + \alpha_2 G(|\phi_2\rangle)$$

$$\Downarrow$$

$$G\left(\sum_{n=1}^d \alpha_n |\phi_n\rangle\right) = \sum_{n=1}^d \alpha_n G(|\phi_n\rangle)$$

ONCE WE KNOW HOW G OPERATES ON THE BASIS VECTORS, THEN WE CAN EASILY CALCULATE THE RESULT FOR ANY OTHER VECTOR OF THE SPACE!

KERNEL

- THE SET OF VECTORS THAT THE OPERATOR G MAPS TO THE ZERO VECTOR, IS CALLED KERNEL OF G ($\text{KER } G$)
- FOR ANY OPERATOR G ON $\mathcal{H}$ THE KERNEL $\text{KER } G$ IS A SUBSPACE OF $\mathcal{H}$
- IF AND ONLY IF $\text{KER } G$ IS THE ZERO SUBSPACE OF $\mathcal{H}$ - IF IT CONTAINS THE ZERO VECTOR AND NO OTHER - THEN G IS ONE-TO-ONE OPERATOR

$$1. |\psi\rangle, |\omega\rangle \in \text{KER } G \quad \alpha |\psi\rangle + b |\omega\rangle \stackrel{?}{\in} \text{KER } G$$

$$G(\alpha |\psi\rangle + b |\omega\rangle) = \alpha G(|\psi\rangle) + b G(|\omega\rangle) \stackrel{||}{=} 0$$

$$2. \text{ IF } \text{KER } G = \emptyset \text{ THEN } G \text{ IS A ONE-TO-ONE OPERATOR}$$

$$|\psi\rangle, |\omega\rangle \quad G(|\psi\rangle) = G(|\omega\rangle) \Rightarrow G(|\psi\rangle) - G(|\omega\rangle) = 0$$

$$|\psi\rangle - |\omega\rangle \in \text{KER } G$$

IF G IS ONE-TO-ONE OPERATOR THEN PROVE THAT $\text{KER } G = \emptyset$

$$\text{ASSUME } |\psi\rangle \stackrel{?}{\neq} 0, G(|\psi\rangle) = 0 \quad G(|\omega\rangle + |\psi\rangle) = G(|\omega\rangle) + G(|\psi\rangle) \stackrel{||}{=} 0$$

$$|\omega\rangle \neq |\omega\rangle + |\psi\rangle$$